

OPERATING SYSTEMS

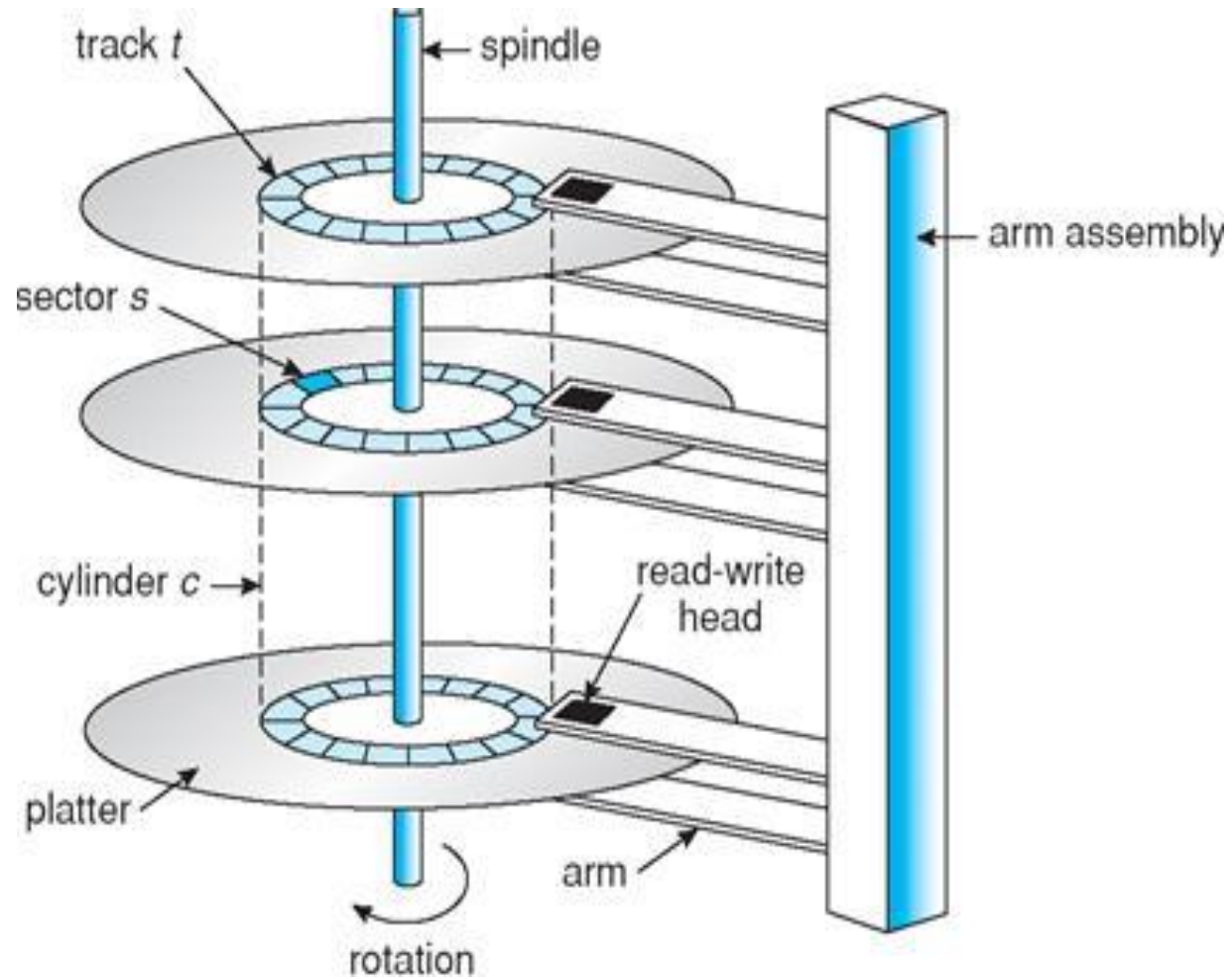
Storage Management

S.Thenmozhi

Department of Computer Applications

OPERATING SYSTEMS

Disk Structure



Magnetic disks provide bulk of secondary storage of modern computers

Drive **rotate** at 60 to 200 times per second

Transfer rate is the rate at which data flow between drive and computer

Positioning time (seek time) is time to move disk arm to desired cylinder

Rotational latency - time to rotate the disk head

Head crash results from disk head making contact with the disk surface

Disks can be **removable**

Disk drives are addressed as large **1-dimensional arrays** of **logical blocks**, where the logical block is the **smallest unit of transfer**

The 1-dimensional array of logical blocks is mapped into the **sectors** of the disk sequentially.

Sector 0 is the first sector of the **first track**

Mapping proceeds in order through that track, then the rest of the tracks in that cylinder and then through the rest of the cylinders from **outermost to innermost**.



THANK YOU

S. Thenmozhi

Department of Computer Applications

thenmozhis@pes.edu

+91 80 6666 3333 Extn 393